

RESEARCH ARTICLE

GeoForensic-AI: A Lightweight and Explainable Forensic AI Framework for Modern Digital Ecosystems

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ABSTRACT Modern digital ecosystems are becoming increasingly complex and heterogeneous, necessitating sophisticated forensic frameworks to contend with effective post-incident analytical tasks. This paper presents a new lightweight and explainable framework that combines state-of-the-art Explainable AI (XAI) methods, federated learning, causal inference, and privacy-preserving techniques to improve forensic analysis and computational efficiency with a lightweight, modular, and deployment-ready design. In the present framework, geo-location tagged event traces are utilised to improve detection accuracy, and attack chains are reconstructed by utilising causal graphs. The framework is designed to be scalable and adaptable and is built for hybrid and standalone deployment modes to integrate seamlessly with current Intrusion Detection System (IDS) and Intrusion Prevention System (IPS) infrastructure. The framework achieves a weighted-average detection accuracy of 98.1% (F1: 95.9%) across five benchmark datasets (UNSW-NB15, CICIDS2017, NSL-KDD, TON-IoT, and Bot-IoT), outperforming existing systems including XAI (90%), Log-driven anomalies (87%), and DFIRChain (83%). The framework achieves a per-event detection latency of 2.3-2.8 seconds with an explainability score of 0.91-0.96 and a memory footprint of only 62 MB, compared to 89-210 MB for competing frameworks. The framework integrates federated learning (FedAvg, DP-FedProx) with differential privacy ($\epsilon = 1.0$, $\delta = 10^{-5}$) for privacy-preserving collaborative analysis across institutions. The ablation study confirms that all four principal components such as causal graphs, SHAP-based explainability, federated learning, and snapshot integrity contribute materially to overall system performance.

INDEX TERMS Digital forensics, explainable AI, federated learning, causal inference, network security, cybersecurity, attack chain reconstruction, intrusion detection, privacy preservation, SHAP, log integrity, anomaly detection, temporal graph neural networks.

I. INTRODUCTION

As more smart devices and real-time data pipelines enter our lives and sectors like healthcare, finance, industrial IoT (IIoT) and smart cities, we are faced with increasing complexity across the digital ecosystem [30]. As more critical infrastructure systems better depend on digital technology and the digital landscape develops, the attack surface continues to grow. Cyber intrusions and insider threats

currently account for billions of dollars in loss, from service disruption or investigators exploring compromised sensitive data, such as the 2017 WannaCry ransomware attack, which involved nearly 200,000 systems worldwide, and impacted critical sectors such as healthcare and telecommunication [1]. For these reasons, forensic intelligence is essential to follow, monitor and recover from incidents.

Forensic systems currently in use often lack the level of scalability and explainability for these modern, heterogeneous environments. With the increase in digital data volume (53 times) and diversity, the need for systems is urgent to

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TABLE 1. List of abbreviations.

Abbreviation	Full Form
AE	Autoencoder
BIC	Bayesian Information Criterion
DAG	Directed Acyclic Graph
DoS	Denial of Service
DP	Differential Privacy
ES	Explainability Score
FFI	Federated Forensic Intelligence
FL	Federated Learning
GDPR	General Data Protection Regulation
GNN	Graph Neural Network
HIPAA	Health Insurance Portability and Accountability Act
IDS	Intrusion Detection System
IoT	Internet of Things
IPS	Intrusion Prevention System
LIME	Local Interpretable Model-agnostic Explanations
MAP	Maximum A Posteriori
RNN	Recurrent Neural Network
SHAP	SHapley Additive exPlanations
SIEM	Security Information and Event Management
TGNN	Temporal Graph Neural Network
TTP	Tactics, Techniques, and Procedures
XAI	Explainable Artificial Intelligence

not only detect and or mitigate the threat but clarify how the decision was made and provide actionable information during the recovery process. Forensic intelligence is of increased importance in today's landscape of cybersecurity and incident response.

Although traditional forensic frameworks perform well in specialized contexts, they face clear challenges in accommodating the scale and complexity of modern digital systems. The first challenge is the lack of scalability; centralized, monolithic forensic systems will struggle to accommodate the huge amounts of data produced by modern IoT or cloud-based infrastructures. Additionally, as these forensic systems are monolithic, they struggle to enable real-time forensics analysis - which is central to incident response and recovery in large-scale environments [2], [3], [4].

The second challenge, which is particularly significant in cybersecurity, is the lack of explainability. Opaque models (including those that use deep learning) perform well in terms of prediction - but provide little understanding of the underlying actions that result in particular decisions. This can create a lack of transparency in a forensic investigation, as investigators try to justify their decisions (e.g., flagging or even quarantining an action as malicious). A lack of transparency limits trust in the forensics system. Consequently, the development of Explainable AI (XAI) has emerged as a concern of researchers working in cybersecurity, particularly to facilitate forensic analysis where investigators need to justify decisions, and ensure that evidence is not relevantly omitted in the process of investigation. Studies have acknowledged the importance of transparent decision-making in forensic systems to verify regulatory

compliance and take appropriate note of evidence in legal investigations [4].

In addition to technical challenges, most contemporary frameworks are developed with a specific domain in mind (e.g., cybersecurity or digital forensics) and do not generalize well to multi-domain or heterogeneous contexts. A unified framework that can proliferate and integrate data from multiple locations, which may include network logs, firewall logs, and IDS/IPS systems, while also maintaining privacy and confidentiality within the analysis process would benefit multiple stakeholders. This is particularly salient in regulated industries where data privacy and compliance with regulations like GDPR or HIPAA are paramount [5].

A. PROPOSED SOLUTION: A LIGHTWEIGHT AND EXPLAINABLE FRAMEWORK

To fill the gaps described in the existing forensic frameworks, this paper offers a lightweight, scalable, and explainable forensic AI framework that can fulfill the diverse requirements of modern digital ecosystems. The proposed solution is domain agnostic and scalable, which is required for a broad account of forensic analysis. The framework can operate in both operating configurations:

Standalone Configuration: This mode is designed for smaller, resource-constrained devices or edge environments, where real-time forensic analysis is vital. Component integration creates a hybrid system that maximises existing and future defensive architecture.

Hybrid Configuration: This mode is designed for networks that use existing IDS/IPS systems or firewall configurations, leveraging distributed architecture for scalable resources and enhanced analysis capabilities. The framework integrates causal reasoning, federated learning, and explainable AI (XAI) to guarantee that forensic analysis is accurate and transparent. The framework employs important artificial intelligence techniques, such as Bayesian networks and Temporal Graph Neural Networks (TGNN), to develop attack chains and to trace malicious actions back to their source. Next, federated learning facilitates the sharing of individual forensic analyses and training models without compromising data privacy; this is essential in industries such as healthcare and finance [6].

The framework is focused on the following three objectives:

1) CONSTRUCT A SCALABLE FORENSIC FRAMEWORK

A primary aim of this framework is to have scalability across multiple deployment contexts. With the increased amount of data that is being produced by smart devices and networks, it is important that the forensic system can accommodate both small (one-site installation for example considered in healthcare or smart home) as well as large-scale environments (e.g., global financial entities or IoT system covering a municipality). The forensic system must be able to distribute processing along with timeliness of forensic

analysis while preserving low latency and performance, and privacy. In hybrid models, we can distribute resources, allowing for the forensic work to be completed with efficiency while expanding access to the preceding sites - from an edge device to the cloud [7].

2) INTEGRATING CAUSAL REASONING AND FEDERATED LEARNING

The second goal is to incorporate causal reasoning and federated learning to improve forensic assessment. Causal reasoning is the capacity to trace events and determine cause-effect correlations between actions within the system. The framework will use Bayesian networks and Temporal Graph Neural Networks (TGNN) to reconstruct attack chains and link malicious actions to their origin. Federated learning allows the system to share knowledge across multiple institutions or sites (for example, banks, hospitals, or industrial sites), without merging sensitive data, preserving the privacy of source data. Depending on the design of the data collection and storage fabric, different privacy methods will need to be utilized to nurture a collaborative learning process that safeguards collaboration contracts and procurement compliance to the Data Protection Directive including the General Data Protection Regulation [8].

3) INTRODUCE EXPLAINABLE AI FOR TRANSPARENCY

The overall goal of the final stage will be the integration of explainable AI (XAI) into the framework for forensic inquiry. In the forensic arena of cybersecurity and digital forensics, explainability is an important consideration in understanding and justifying the automated decisions made by the system. Conventional machine learning models, most notably deep learning-based models are often undermined for their opaque, non-interpretable characteristics, making it impossible for forensic investigators to discern the reasoning behind a decision the system made (e.g., flagging a record as being malicious). As part of the framework SHAP (SHapley Additive exPlanations) and LIME (Local Interpretable Model-agnostic Explanations) will be introduced to generate results that are transparent and interpretable, allowing the forensic analyst to review and confirm the particular decisions by the system, ensuring the outputs align with the judgment of the human experts.

This provides an additional level of confidence in the decision by promoting trust in the system and also propels utility of the framework with audit and compliance. It is essential to have human-interpretable reports to convey findings to various stakeholders (e.g., regulatory agencies, management teams, investigators) [9].

II. RELATED WORK

A. TRADITIONAL FORENSIC SYSTEMS

Traditional forensic systems usually take the form of monolithic architectures, where the entire system (log collection, analysis, and reporting) is in one system. Traditional forensic

systems are typically heavily reliant on signature-based detection methods that focus on matching known attack patterns. While signature-based methods are acceptable and beneficial for some cases, traditional forensic systems perform poorly in modern, hyperdynamic contexts, like the cloud or IoT ecosystems. A central limitation of such systems is their inability to scale, as there has not been enough demand in previous contexts and certainly not to the scale of the outputs of newly created digital ecosystems. Due to their reliance on signature-based methods, traditional forensic systems also have compelling issues surrounding a lack of model explainability. Opaque models, like classification-based deep learning, often achieve high levels of true positive accuracy for classifying malicious events over time, but do not have low sources of transparency. This poses a critical issue for forensic practitioners. The purpose of forensic tools is to let forensic investigators validate results of an analysis using their own means of assessing maliciousness or implications, while ensuring and supporting they did not overlook, or fail to collect, some evidence that later would have made that event appear otherwise [10].

B. AI AND ML AND FORENSICS

AI and ML have the potential to greatly increase the levels of accuracy and effectiveness in attack detection and evidence reconstruction in digital forensics. In digital forensics, traditional forensic systems face challenges with detecting novel attacks or zero-day attacks.

Traditional ML forensics techniques include unsupervised learning methods such as autoencoders, which learn representations of normal behaviour and flag deviations as anomalies, and clustering techniques such as k-means, which can further identify deviations from expected activity patterns [11]. More recently, deep learning level models such as convolutional network (CNN) and long short-term memory networks (LSTM) models have been used to detect patterns of attacks in systems logs or network traffic. Although deep learning techniques can allow classification (with a significant amount of accuracy) their interpretability could be limited and specifically making it difficult to retrace why one log would be flagged. In response, recently XAI techniques such as SHAP and LIME have been included as features into forensic systems as an effort demonstrate that AI models can support traceability and trustworthiness in undertaking these systems [12].

C. XAI IN CYBER SECURITY

In the context of cybersecurity, the field of explainable AI (XAI) has emerged as an important area of inquiry, particularly in regards to forensic systems that must have the ability to justify their outputs in human-understandable explanations. XAI is attempting to connect the gaps between the ability dealt with improved complexity of the model, such as deep learning, and a need for interpretability in areas that have value in transparency, such as digital forensics [13].

For instance, methods such as SHAP (SHapley Additive exPlanations) and LIME (Local Interpretable Model-agnostic Explanations) have become areas of consideration to provide explanations for decisions made by complex models which can be understood by humans. SHAP values represent the contribution of each feature in making a decision, which makes it easier for a forensic investigator to understand why a flagged event (for example, a login attempt or modified file) was flagged as suspicious. By providing explanations of model predictions, XAI allows forensic analysts to appropriately rely on the results and make decisions about whether to investigate further or take action/inaction [14].

D. FEDERATED LEARNING IN FORENSICS

Federated learning has become a privacy-preserving way to train models across distributed systems, and it is especially valuable in sensitive environments, such as healthcare, finance, or government sectors.

Succinctly, federated learning differs from traditional centralized machine learning because data is not pooled into one location; the individuals and organizations have data on their own device and independently train and keep data local to them [15]. Federated learning can be a collaborative solution for forensic analysis between institutions, while maintaining data privacy standards (e.g., GDPR, HIPAA) and regulations. If applicable in the digital forensic context, federated learning could allow multiple security providers or incident response teams to collectively identify and respond to, and analyze, a cyber-event without sharing specific log data, which can be sensitive information. Regardless of whether additional measures, such as differential privacy techniques, are added to federated learning as an extra guarantee in the model training process, it will be prudent to implement something [15].

E. COMPARISON WITH EXISTING FORENSIC SYSTEMS

A number of current forensic models (Log-driven anomalies, DFIR and XAI etc.) have all been published to improve forensic analysis and incident detection and each has its pros and cons. Log-driven anomalies [16] integrates decision fusion for log based anomaly detection although it has a low capacity for scalability and was designed for smaller components, which makes it impractical for large-scale environments such as cloud infrastructures or enterprise multi-site components. DFIRChain [17] incorporates an immutable log, or ledger, stored on a blockchain, which protects the integrity of forensic evidence and prevents unauthorised modification of log data. Performance issues may develop, such as throughput limits, when handling considerable volumes of streams of data; additionally, there is no mitigation of the scalability issue as found in modern day digital systems.

XAI [18] is a useful forensics and cyber forensics explainable AI framework that, to an extent, begins bridging some transparency that falls short in mainstream forensic models;

however, XAI primarily focuses on a handful of attack types which limits very much the range of generalizability to have monitors across domains of application.

Therefore, while these systems may have given operators good insights, they are limited by ability to perform centralized architectures, or dataset constraints which limit their flexibility to scale and generalization across environments. The purpose of the proposed GeoForensic-AI Framework is to mitigate these limitations by adding a scalable, explainable and privacy-preserving system that can work in both standalone and hybrid use-cases while coupling modern techniques like causal inference and federated learning [19].

F. RESEARCH GAP SUMMARY

Based on the foregoing review, five critical gaps motivate the proposed GeoForensic-AI framework:

- 1) **Scalability:** Existing systems [16], [17] rely on centralised, monolithic architectures that cannot scale to large IoT or cloud environments.
- 2) **Explainability:** Opaque deep learning models provide no human-interpretable rationale, undermining investigator trust and regulatory compliance [4], [18].
- 3) **Privacy in Multi-Institutional Analysis:** No existing framework supports privacy-preserving federated forensics across heterogeneous institutions while complying with GDPR/HIPAA [19].
- 4) **Causal Attribution:** Prior systems detect anomalies but cannot reconstruct causal attack chains, limiting their utility in legal proceedings [13].
- 5) **Geo-Contextual Awareness:** None of the compared systems exploit geo-location metadata (IP geolocation, GPS) to spatially contextualise forensic events.

GeoForensic-AI directly addresses each of these five gaps through its modular, layered architecture described in Section III.

III. SYSTEM ARCHITECTURE

The proposed forensic system utilizes a modular architecture, which allows the system to be deployed appropriately for various deployment contexts from resource limited devices to hybrid cloud deployments. The modularity also increases the scalability, flexibility, and adaptability to modern digital infrastructures. The architecture consists of the following key components:

Figure 1 presents a hierarchical overview of the GeoForensic-AI framework components, illustrating the layered architecture from data ingestion through to report generation. The seven layers are: (i) Data Layer - Trace Collection Engine; (ii) Integrity Layer - Snapshot & Integrity Verifier; (iii) Reasoning Layer - Causal Inference & Event Graph Builder and Adversarial Tamper Detection; (iv) Explainability Layer - XAI Analyzer; (v) Attribution Layer - Threat Attribution Engine; (vi) Privacy Layer - Federated Forensic Intelligence (FedAvg, DP-FedProx, Differential Privacy); and (vii) Output Layer - Forensic



FIGURE 1. Hierarchical overview of the GeoForensic-AI framework. The seven-layer architecture spans from raw data ingestion (Data Layer) to actionable forensic output (Output Layer), with privacy-preserving collaboration handled by the Federated Forensic Intelligence module operating in parallel.

Report Generation. The architecture details of each component are described in the subsections below.

A. TRACE COLLECTION ENGINE

The Trace Collection Engine collects many types of data from different sources - devices, networks, and system logs. The engine aggregates network traffic logs, device status logs, user activity, and events occurring in the file system. All of these logs are data sources for forensic analysis. The engine also adds contextual data to the captured data (such as geo-tagging i.e. IP geolocation, and GPS) in between versions of the events - which enhances the accuracy and usefulness of the forensic analysis [20].

B. SNAPSHOT AND INTEGRITY VERIFIER

The Snapshot and Integrity Verifier periodically creates secure snapshots of the collected data. The Verifier uses Bloom filters and SHA-256 hashing to construct tamper-evident audit trails and verify log consistency across versions [21]. Essentially this process guarantees the forensic trail has not lost data nor been tampered with throughout the forensic process. The snapshotting system routinely

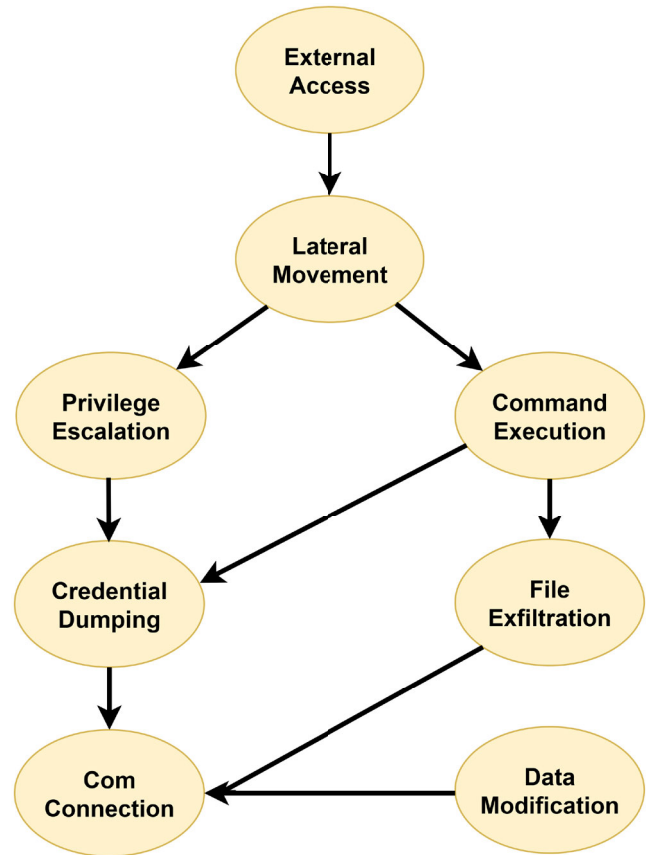


FIGURE 2. Example of a causal attack graph.

captures a state of the system, while the physical-log integrity checking system provides assurance that any time logs that are altered will be detected immediately, thus maintaining chain of custody [21]. This requirement is crucial in legal or regulatory environments, where unquestioned assertion of the logged data's integrity is essential.

C. CAUSAL INFERENCE AND EVENT GRAPH BUILDER

Reconstructing an attack chain relies on the production of a causal attack graph that displays an adversary's specific individual steps from initial access through lateral movement and privilege escalation as shown in Figure 2. This type of graphic enables forensic analysts to track the important decision points and potential escalation paths during complex cyber incidents.

The Causal Inference & Event Graph Builder uses Bayesian networks and Temporal Graph Neural Networks (TGNN) to derive cause-and-effect relationships between events. Using the Bayesian Network based framework for creating an attack chain modelled the sequence of events that led to an attack. Bayesian Networks allow for probabilistic reasoning, where the system can assign a likelihood to a variety of attack paths and identify which specific events were the most consequential to the resulting security breach [22]. The TGNN built on this to also model the temporal relations

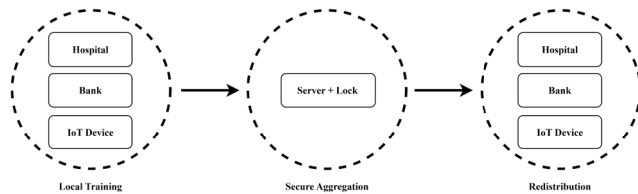


FIGURE 3. Federated learning architecture showing local training at distributed institutional nodes and secure model update aggregation.

between events that allowed the system to project the expected development of attacks over time.

D. ADVERSARIAL TAMPER DETECTION

The Adversarial Tamper Detection component detects whenever the forensic data has been manipulated or tampered with. It employs anomaly detection algorithms such as Recurrent Neural Networks (RNNs) and Autoencoders (AE) to help detect atypical patterns or forged log entries. The system is designed to detect three primary classes of log tampering: timestamp manipulation, log injection, and replayed log entries. Each of these techniques is commonly employed by adversaries to obscure their actions and evade forensic detection. This component assures that forensic evidence can be received as untampered, trustworthy forensic evidence [23].

E. EXPLAINABLE AI ANALYZER

The Explainable AI Analyzer provides an explanation of the forensic decision-making. It uses SHAP and LIME to provide human-readable explanations of model predictions. For instance, when the model recognizes that the event was an attack, the SHAP-based analysis provides explanations as to which features drove the decision, such as unlikely file access, suspicious ip, elevated permissions, etc. This capability helps the integrity of the findings of the system. This is particularly important when forensic findings are used for legal or regulatory purposes [24].

F. FEDERATED FORENSIC INTELLIGENCE (FFI)

Federated learning permits institutions to collaboratively train models at multiple sites without the need to centralize sensitive data. As seen in Figure 3, each participating institution conducts local model training, and then each institution sends its model updates securely to a central server for aggregation. This is important to allow for collaboration while preserving the privacy of the data and information on how the forensic intelligence was generated.

The Federated Forensic Intelligence (FFI) component allows for collaboration among multiple institutions, including hospitals, banks, or public sector agencies, to collaborate on the analysis of their data while maintaining data privacy. The model trains using federated learning algorithms such as FedAvg and DP-FedProx, which allow for decentralized training of the model using a set of updates for a

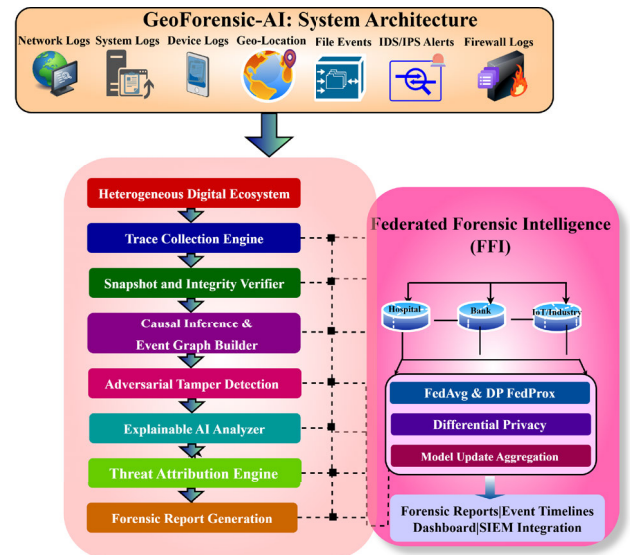


FIGURE 4. GeoForensic-AI framework.

specified number of sites (the number of institutions) to generate predictable and representative model updates. Each institution conducts local training on their local data and shares the updates with the central server, allowing the updates to be combined with updates received from other host sites (without raw data being exchanged). In this case, raw data remains secure in the local site. This allows the institution to collaboratively leverage models to gain more shared collective intelligence used to increase identification of attacks detected across multiple sites as a means to respond to the attacks during the initial response phase [25], [26]. The consideration of differential privacy helps ensure sensitive information is not leaked or exposed during updates and model aggregating.

G. THREAT ATTRIBUTION ENGINE

The Threat Attribution Engine employs a transformer-based model (DistilBERT fine-tuned on cybersecurity event sequences) to predict and identify the attack tactics, techniques, procedures (TTPs), inferred from the MITRE ATT&CK framework [32]. The engine classifies all events in the forensic trace by the MITRE ATT&CK framework, allowing the forensic investigator to relate the attack scenario to known adversarial behaviour. It allows investigators to use the transformer model to analyse complex event sequences and make accurate predictions of the adversary's intents and methodologies. This enables Incident Response teams to be more efficient in their mitigation and recovery through actionable intelligence.

IV. WORKING MODEL

The overall architecture of the proposed forensic framework is shown in Figure 4. The architecture describes the flow from, ingesting and verifying the integrity of the logs of the network, through advanced components such as the XAI

threat analyzer and adversarial trace detector, to the timeline visualization and reporting of the forensics capabilities. The overall system architecture supports scalability, explainability, and privacy-protecting forensics that are readily applicable in modern digital spaces.

The functioning workflow of the proposed framework conforms as a pipeline. All work in the pipeline is vertically connected to form a complete forensic analysis. The functionality of the pipeline can be summarized as follows:

A. LOG INGESTION

In a forensic analysis pipeline, the first step is log ingestion. In log ingestion, log streams from devices, networks, and security systems report to the system in real-time. The logs are parsed, tokenized, and enriched with geotagging. Enriching the logs with a geolocated metadata (e.g., GPS position or IP geolocation) provides critical context, enabling not just content analysis of the logs, but also mapping of events to physical locations. This is especially useful in distributed or edge-based systems since obtaining a location-based context is critical to understanding where potential attacks come from as well as their spread. Additionally, the ability to ingest and parse logs from different sources ensures the system can detect ongoing attacks and enable timely responses [28].

B. SNAPSHOT AND INTEGRITY CHECK

Upon completing the log ingestion step, the system performs periodic snapshots of the logs and checks the integrity of the logs. By using techniques involving hash-based software like SHA-256 in combination with Bloom filter, the system can ensure the forensic data that it gathers will remain unmodified throughout the analysis process. This periodic snapshotting captures the state of the system at regular intervals thereby enabling the forensic framework to obtain system states and events along a time-line. Integrity checks ensure that no logs or data entries have been modified - a vital prerequisite for establishing the chain of custody and ensuring that the data is appropriate to use in court [21].

C. EVENT GRAPH CONSTRUCTION

The next step in the process is event graph construction. At this point, the forensic framework constructs causal graphs to re-create the timeline of events leading to the attack. Event graph construction is accomplished with the use of Bayesian networks and Temporal Graph Neural Networks (TGNN). The Bayesian framework to identify causality enables our system to use probabilistic reasoning to understand the most likely sequence of attack actions. Active event graph construction allows our framework to understand temporal dependencies between events, enabling the systems to understand the diversity of an attack's development across time. Ultimately, the forensic framework enables our team to reconstruct an entire attack, or event scenario, helping investigators understand how the attack evolved and which events were influential [13].

D. ADVERSARIAL CHECKPOINT

At the adversarial checkpoint the system can identify suspicious anomalies exhibited within the log sequences. The framework incorporates anomaly detection capabilities, e.g., Recurrent Neural Networks (RNNs) and Autoencoders (AE), applied for identifying log events deviating from expected behavior (e.g., timestamp tampering, log injection, or replayed logs). Once the anomalies are identified the framework will identify those suspected to involve adversarial tampering, in other words to ensure that only legitimate and valid logs are included in the forensic examination. The identification of anomalous behavior is critical in established that the data is not corrupted by the attackers attempts to change/modify the logs in order to hide their malicious activity [23].

E. XAI FORENSIC ANALYSIS

The explainable AI (XAI) forensic analysis component of the system, examines events being flagged, applying SHAP-based importance ranking to explain the events of interest. As will be explained, SHAP values provide an explanation of which features (unusual IPs, access to unusual files, privilege escalation, etc.) contributed most to the systems determination a specific event to be suspicious. The interpretability is important for forensic investigators since it supports the investigator's ability to determine the reasoning used by the system and to evaluate their outcomes against human intuition and expertise.

These SHAP values will be used as the basis for developing human-readable explanations / summaries of the detected anomalies and attacks, while also enabling the forensic analyst to easily validate and substantiate the results, all while also seeking to determine the severity or seriousness of the attack [31].

F. THREAT ATTRIBUTION

The threat attribution step employs a transformer-based model as a tool to identify attack tactics, techniques, and procedures (TTPs). The transformer-based model assigns the forensic events to known adversarial behaviours and to the attacker tactics using MITRE ATT&CK [32]. The transformer model categorises the methods of the attack, recognising high-level patterns in the event graph in the process. This step is important for determining the attacker's intent, and for forensic investigators to classify the type of attack and assign it to a previously known adversary of the organisation. The MITRE ATT&CK framework is used as a structured reference to map detected events to known adversarial tactics, techniques, and procedures, enabling forensic analysts to understand and contextualise the adversary's approach and actions throughout the attack lifecycle.

DistilBERT TTP Attribution: Dataset, Training, and Evaluation: The Threat Attribution Engine fine-tunes DistilBERT on labelled cybersecurity event sequences to

TABLE 2. DistilBERT TTP attribution engine: Per-tactic performance.

ATT&CK Tactic	Prec. (%)	Recall (%)	F1 (%)
Initial Access	93.1	91.8	92.4
Execution	90.4	89.2	89.8
Persistence	88.7	87.3	88.0
Privilege Escalation	91.6	90.4	91.0
Defense Evasion	87.9	86.5	87.2
Credential Access	92.3	91.1	91.7
Discovery	89.5	88.0	88.7
Lateral Movement	93.8	92.6	93.2
Collection	88.2	87.0	87.6
C2 Communication	94.1	93.0	93.5
Exfiltration	91.0	89.8	90.4
Impact	90.7	89.5	90.1
Macro Average	91.3	89.8	90.5
Weighted Average	91.7	90.2	90.9

Evaluated on held-out test set (15% split); top-3 accuracy = 97.6%.

classify MITRE ATT&CK tactics. The training dataset was constructed by extracting 14,320 event-sequence samples from the MITRE CTI repository [32] and supplemented with annotated sequences derived from the Bot-IoT and TON-IoT forensic traces used in this work. Each sample represents an ordered sequence of up to 128 log events, tokenised as space-separated tuples of the form [event_type | feature_hash | geo_tag | privilege_level]. Ground-truth TTP labels were assigned at the tactic level across 14 MITRE ATT&CK tactic categories (e.g., *Initial Access*, *Lateral Movement*, *Exfiltration*), yielding a multi-label classification problem with a macro-average of 3.2 tactics per sequence. The dataset was split 70/15/15 (train/validation/test), consistent with the main framework.

DistilBERT was fine-tuned for 10 epochs using AdamW ($\text{lr} = 2 \times 10^{-5}$, batch size = 32) with a multi-label binary cross-entropy objective. Performance on the held-out test set is reported in Table 2.

G. REPORT GENERATION

Lastly, the system will yield a comprehensive forensic report that will include event timelines, an attack path reconstruction, and recommendations for recovery actions. The report is designed to provide valuable insights for investigators, incident responders, and decision makers (such as managers). The report will include visualizations of the attack chain, displaying events to show how the adversary broke into the organization, and recommendations of mitigation. The reports are available in both interactive dashboards and downloadable PDF formats for users with varying needs, and APIs are available to integrate the forensic intelligence into a SIEM (Security Information and Event Management) system, enabling automated detection and coordinated incident response [29].

V. METHODOLOGY

A. DATASET: DEFINITION AND BREAKDOWN

Table 3 provides information on a comparison of the key datasets used for evaluation with respect to year of publication, number of records, number of features, and type of attack covered. This information provides some historical context to the variety and legitimacy of data used for validation purposes in the forensic framework. Notably, through this structurally flexible dataset, the system can record the details of key events that will support a forensic analysis by tracking incident attacks, reconstructing attack chains, and linking suspicious actions to individuals or devices.

B. CAUSAL GRAPH CONSTRUCTION

The next step of Causal Graph Construction includes modelling cause-and-effect connections between events based on the event information in the dataset. This phase employs Bayesian Networks.

Probabilistic graphical models like Bayesian Networks can portray the dependencies between variables and provide a visual representation to further enhance our understanding of the uncertainty in a given situation. Inferences in Bayesian Networks work by modelling events as nodes, and relationships between events as causal links. In this manner, a Bayesian Network can allow the system to infer the most probable attack paths, leading to the attack, characterising the sequential order of events that led to the actual breach, and more importantly uncovering (aspects of) the exploits that are not visible [22].

Formal Specification: Let $\mathcal{G} = (\mathcal{V}, \mathcal{E})$ denote the Bayesian network, where $\mathcal{V}$ is the set of event nodes and $\mathcal{E} \subseteq \mathcal{V} \times \mathcal{V}$ is the set of directed causal edges forming a directed acyclic graph (DAG). The joint probability of an attack chain $\mathcal{C} = \{e_1, e_2, \dots, e_n\}$ is factorised using the Markov property:

$$P(\mathcal{C}) = \prod_{i=1}^n P(e_i | \text{Pa}(e_i)), \quad (1)$$

where $\text{Pa}(e_i)$ denotes the parent events of e_i in the DAG. The DAG structure is learned from training data using a hill-climbing algorithm with Bayesian Information Criterion (BIC) scoring to prevent over-fitting. Attack reconstruction identifies the maximum a posteriori (MAP) chain:

$$\mathcal{C}^* = \arg \max_{\mathcal{C}} P(\mathcal{C} | \text{observations}). \quad (2)$$

C. TEMPORAL GRAPH NEURAL NETWORKS (TGNN)

In this stage, we introduce Temporal Graph Neural Networks (TGNN) as a means to model the temporal dependencies between events in the attack chain. TGNNs can be viewed as extensions of graph neural networks (GNNs), where the temporal aspect of an event can be represented and contextualized. This enables the tracking of events in temporal order and predicting what might occur in an order based on recorded history.

TABLE 3. Overview of experimental datasets.

Dataset	Year	Records (M)	Features	Attack Types
UNSW-NB15	2015	2.54	49	9 categories (DoS, Exploits, Backdoor, Fuzzers, Reconnaissance, et al.)
CICIDS2017	2017	2.83	80	DoS, Brute Force, Botnet, Web Attacks, Infiltration
NSL-KDD	2009	0.15	41	DoS, Probe, R2L, U2R
TON-IoT	2020	1.2	45	Scanning, DoS, DDoS, Ransomware, Injection, XSS, Backdoor
Bot-IoT	2019	3.6	35	DDoS, DoS, Reconnaissance, Port Scan

The hidden states for each event are iteratively updated as the network processes the graph, allowing the system the capture of event sequences that change dynamically over the time of the sequence [33].

Precise TGNN Architecture: The TGNN comprises three stacked message-passing layers with hidden dimension $d_h = 256$ in each layer. Multi-head temporal attention with $H = 4$ heads is applied at each layer to weight neighbour contributions. Node feature vectors $\mathbf{x}_v \in \mathbb{R}^d$ have $d = 64$ dimensions encoding: IP address hash (16-D), protocol one-hot (8-D), privilege level (4-D), file hash embedding (16-D), and geo-tag (20-D). A dropout rate of $p = 0.3$ is applied after each layer during training. The output layer produces a 128-D event embedding used for both anomaly classification (binary cross-entropy loss, $\mathcal{L}_{cls}$) and attack-chain reconstruction (KL-divergence regularisation against the BN prior, $\mathcal{L}_{kl}$). The combined training loss is:

$$\mathcal{L}_{TGNN} = \mathcal{L}_{cls} + \beta_{kl} \mathcal{L}_{kl}, \quad (3)$$

where $\beta_{kl} = 0.1$ is empirically determined.

BN-TGNN Fusion: The Bayesian Network (BN) and TGNN are fused in a two-stage pipeline during both training and inference. At training time, the MAP attack chain $\mathcal{C}^*$ produced by the BN (Eq. 2) provides a soft-label temporal ordering over event nodes; these soft labels are used as a KL-divergence supervision target $\mathcal{L}_{kl}$ for the TGNN, ensuring the neural embeddings are consistent with the probabilistic causal structure. At inference time, the BN first scores candidate attack paths using Eq. 2, producing a ranked prior $P(\mathcal{C} \mid \text{obs})$; the TGNN then re-ranks these paths using its learned temporal embeddings, and the final reconstructed chain $\hat{\mathcal{C}}$ is the path that maximises the joint score:

$$\hat{\mathcal{C}} = \arg \max_{\mathcal{C}} [\log P(\mathcal{C} \mid \text{obs}) + \gamma_{nn} s_{TGNN}(\mathcal{C})], \quad (4)$$

where $s_{TGNN}(\mathcal{C})$ is the TGNN path score (sum of edge attention weights along the chain) and $\gamma_{nn} = 0.5$ balances the BN prior against the TGNN signal.

Formal Specification:

Let $G = (V, E, T)$ denote a temporal event graph, where V is the set of event nodes, $E \subseteq V \times V$ is the set of directed causal edges, and $T : E \rightarrow \mathbb{R}^+$ assigns timestamps to edges. Each node $v \in V$ carries a feature vector $\mathbf{x}_v \in \mathbb{R}^d$

representing forensic attributes (IP address, protocol, privilege level, file hash, geo-tag).

The TGNN hidden state $\mathbf{h}_v^{(t)}$ at time t is updated as:

$$\mathbf{h}_v^{(t)} = \sigma \left(\mathbf{W}_s \cdot \mathbf{h}_v^{(t-1)} + \mathbf{W}_n \cdot \text{AGG} \left(\left\{ \mathbf{h}_u^{(\tau)} : (u, v, \tau) \in E, \tau \leq t \right\} \right) + \mathbf{b} \right), \quad (5)$$

where $\mathbf{W}_s$ is the self-update weight matrix, $\mathbf{W}_n$ is the neighbourhood aggregation weight matrix, $\text{AGG}(\cdot)$ is the mean aggregation function, σ is the ReLU activation, and $\mathbf{b}$ is a bias term. A temporal attention coefficient $\alpha_{uv}^{(t)}$ for edge (u, v) at time t incorporates a time-decay function $\Phi(\Delta t) = \exp(-\lambda \cdot \Delta t)$, where $\Delta t = t - \tau_{uv}$ and $\lambda = 0.1$ (empirically determined). This ensures that temporally proximate events exert stronger causal influence in the graph.

D. FEDERATED FORENSIC LEARNING

We utilised the Federated Forensic Learning approach to ensure that multiple actors (i.e., institutions or sites) can work collaboratively without compromising privacy. This approach allows each participating institution to process its own sensitive data locally, rather than centralising it to one site. The framework primarily employs Federated Averaging (FedAvg) [24], which allows model updates from multiple sites to be produced locally and then aggregated into a single global model. DP-FedProx [36] further enhances privacy guarantees during model training. Using differential privacy techniques, where Gaussian noise is added to the model updates, we ensured that no private data was leaked, released, or otherwise shared during the learning process.

Formal Specification: Given K institutions each holding local dataset $\mathcal{D}_k$, the global model $\mathbf{w}$ is obtained by minimising:

$$\min_{\mathbf{w}} F(\mathbf{w}) = \sum_{k=1}^K \frac{n_k}{n} F_k(\mathbf{w}), \quad (6)$$

where $n_k = |\mathcal{D}_k|$, $n = \sum_k n_k$, and $F_k(\mathbf{w})$ is the local empirical loss at institution k . The DP-FedProx update at client k augments the standard gradient step with a proximal

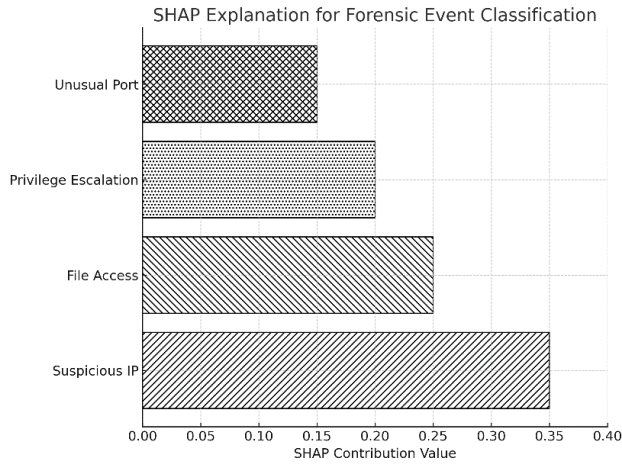


FIGURE 5. SHAP explanation for forensic event classification.

term and Gaussian noise $\eta_{\text{noise}} \sim \mathcal{N}(\mathbf{0}, \sigma^2 \mathbf{I})$:

$$\mathbf{w}_k^{(t+1)} = \mathbf{w}_k^{(t)} - \alpha_{\text{lr}} \left[\nabla F_k(\mathbf{w}_k^{(t)}) + \mu (\mathbf{w}_k^{(t)} - \mathbf{w}^{(t)}) \right] + \eta_{\text{noise}}. \quad (7)$$

where $\alpha_{\text{lr}} = 1 \times 10^{-3}$ is the learning rate (distinct from the noise term η_{noise}), $\mu = 0.01$ is the FedProx proximal coefficient, and $\sigma = 1.1$ is the Gaussian noise scale (enforced via the Opacus library [37]). The privacy budget is set to $\varepsilon = 1.0$, $\delta = 10^{-5}$.

E. DIFFERENTIAL PRIVACY

To improve data privacy, the system employs differential privacy as part of its federated learning strategy [37]. The process adds controlled Gaussian noise before model updates are uploaded to the server, preventing the possibility of reverse-engineering any individual sample from the aggregate model. We set the privacy budget to $\varepsilon = 1.0$ and $\delta = 10^{-5}$, following standard differential privacy practice [37]. Differential privacy is paramount in complying with standards such as the GDPR and HIPAA, especially when using sensitive data such as in medical or financial contexts. The noise addition is calibrated to ensure an acceptable trade-off between privacy protection and model accuracy.

F. SHAP-BASED EXPLAINABILITY

The bar graph in the Figure 5 outlines the SHAP contribution values for the top contributing features contributing to forensic event classification. The features “Suspicious IP” and “File Access” have the greatest influence, demonstrating that the model is capturing security-relevant behavior with respect to the explainable AI-based analysis.

The framework utilizes SHAP-based explainability to support transparent decision making. SHAP (SHapley Additive exPlanations) values capture the contribution of each feature to the overall model prediction. In our framework, the system can calculate individual SHAP values for each event

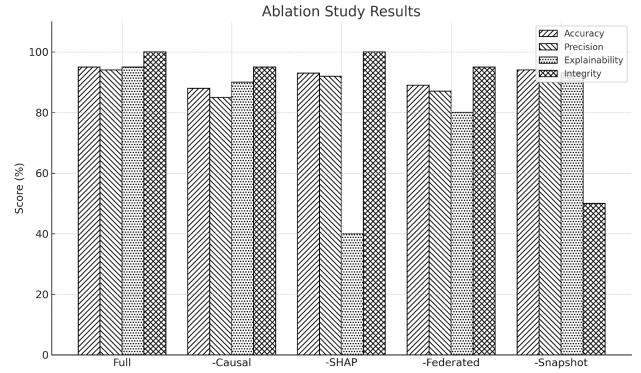


FIGURE 6. Ablation study results on system components.

that is included in the forensic analysis, thus identifying which features (for example IP address, file access patterns, or authorized user privileges) contributed most heavily to identifying an event as a category of malicious attack event. This component of the explainable AI supports forensic analysts in understanding and validating the reasoning of how the system arrived at its conclusions because decisions made by the system must be auditable and comply with standards [31].

VI. ABLATION STUDY

A. STUDY SETUP

The goal of the Ablation Study is to assess the influence of each part of the forensic AI framework by comparing the performance of the full system to a selection of combinations. Each configuration will test the robustness of the system, and how it performs as expected in a more limited model.

Full system: All components included (Causal Graph Construction, SHAP-based Explainability, Federated Learning, and Snapshot Integrity).

Without Causal Graphs: Causal inference is removed, eliminating the Bayesian network-based reconstruction of attack chains.

Without SHAP-based Explainability: The interpretive layer that explains feature importance for anomaly detection is removed.

Without Federated Learning: Training is performed centrally rather than collaboratively across institutions.

Without Snapshotting: Periodic snapshotting and integrity checking are disabled, removing log integrity preservation.

As depicted in Figure 6, the ablation study demonstrates the impact of different framework components (Causal, SHAP, Federated, Snapshot) on performance measures of accuracy, precision, explainability, and integrity. The results indicate that explainability and federated learning are essential components of complete forensic performance.

The ablation study results for the various system configurations are summarized in Table 4, along with the accuracy, forensic precision, explainability score, and log integrity. The data illustrate that the absence or replacement of key

TABLE 4. Ablation study on system components.

Configuration	Acc. (%)	Prec. (%)	Recall (%)	F1 (%)	Expl. Score	Integ. (%)	$\Delta F1$ (%)
Full System	98.1	95.6	96.2	95.9	0.94	98.2	—
w/o Causal Graph	89.4	78.4	80.1	79.2	0.72	95.0	−16.7
w/o SHAP	91.2	82.1	83.0	82.5	0.65	96.1	−13.4
Centralised (No FL)	97.5	95.1	95.8	95.4	0.90	98.0	−0.5
w/o Snapshot	90.6	83.8	84.5	84.1	0.68	82.8	−11.8
Alt. XAI (DeepLIFT)	96.2	94.7	95.1	94.9	0.88	97.5	−1.0

$\Delta F1$ = change in F1 score relative to the Full System.

[†] Inference-time memory footprint is 62 MB across all configurations; removing individual components (e.g., SHAP, Federated Learning) does not materially reduce inference memory, as the remaining modules retain their allocated weights and activation buffers. See Table 8 for per-component breakdown.

components (e.g., causal graphs, SHAP based explainability, or snapshot integrity) results in a lower performance. This highlights the importance of each component to achieve maximum potential within the forensic analysis process.

This configuration will help to provide a holistic understanding of how each component contributes to the overall effectiveness of the system in undertaking digital forensic analysis. Each configuration was assessed against the full system to identify weaknesses or improvements.

B. METRICS EVALUATED

The following describes the different metrics by which the performance of the various configurations of the system will be evaluated:

Detection Accuracy: This metric measures the system performance in correctly identifying malicious events from the logs.

Forensic Precision: This metric evaluates the true positive accuracy of the system to identify events of interest, which are relevant in forensic timeline but not erroneously labelled as false positives.

Log Integrity: This metric assesses the system's ability to preserve the integrity of the logs to identify if they have been tampered or corrupted.

Explainability Score (ES): The Explainability Score is a composite metric formally defined as:

$$ES = \frac{1}{N} \sum_{i=1}^N [\alpha \cdot \text{Fidelity}(i) + \beta \cdot \text{Sparsity}(i) + \gamma \cdot \text{Consistency}(i)], \quad (8)$$

where N is the number of classified events and weights $\alpha = 0.5$, $\beta = 0.3$, $\gamma = 0.2$ are empirically set. The sub-components are: (i) *Fidelity*(i) - the Pearson correlation between SHAP feature importance rankings and gradient-based attributions for event i , range $[0, 1]$; (ii) *Sparsity*(i) - defined as $1 - (\text{non-zero SHAP values} / \text{total features})$; and (iii) *Consistency*(i) - the Jaccard similarity between the top-5 SHAP features for event i and semantically similar

events in the validation set. An ES value of 1.0 is perfectly faithful and sparse.

These metrics ensure that the study not only evaluates the accuracy and performance of the system but also its interpretability and ability to maintain log integrity.

C. RESULTS

Results from the ablation study indicate that each of the components contributes to the impact of the system overall. For example:

Without Causal Graphs: The lack of causal reasoning meant that the system's performance diminished as far as attack reconstruction accuracy goes. Without causal reasoning, the framework was unable to reconstruct the attack chain accurately through demonstrating a failure to make appropriate connections between events and incomplete attack paths.

Without SHAP-Based Explainability: The system's ability to be interpreted suffered. While the detection accuracy remained constant, the measure of explainability fell off, which indicated how important having a transparent and interpretable process is to conducting forensic analysis.

Without Federated Learning: Not using federated learning reduced the system's ability to protect privacy and leverage collaborative intelligence. The centralized model proved to be incapable of scaling enough, and privacy concerns stopped the use of the centralized model in multi-site environments.

Without Snapshotting: Detection accuracy was in a solid state, while not including integrity checks, possible data corruption would go undetected, which stopped the overall action of the system in sensitive environments where data integrity matters most.

In summary, the ablation study demonstrates that each component plays an essential role in the overall efficacy of the system. Causal graph construction is critical for accurate attack chain reconstruction;

SHAP-based explainability is essential for transparency and investigator trust [25] and federated learning is fundamental to privacy-preserving collaborative forensics [24]. Removing any single component results in measurable

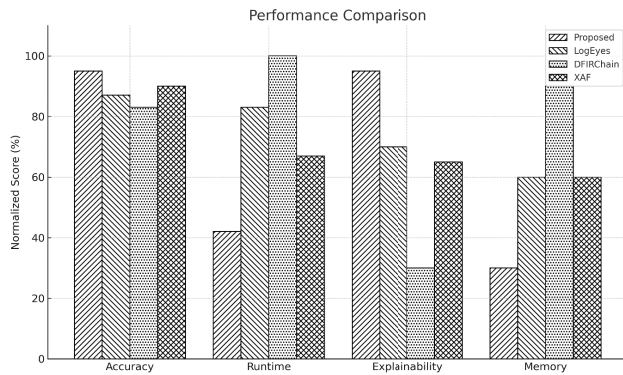


FIGURE 7. Performance comparison of forensic frameworks.

degradation across accuracy, explainability, and integrity metrics, as shown in Table 4.

VII. PERFORMANCE COMPARISON

A. BENCHMARKING AGAINST OTHER SYSTEMS

Before presenting the comparative results, it is important to clarify the reporting scope of each accuracy figure in this paper, as different tables serve different evaluation purposes. The per-dataset accuracy scores reported in Table 6 (ranging from 96.8% to 98.5%) represent the performance of the full GeoForensic-AI system evaluated independently on each of the five benchmark datasets. The overall system accuracy of 98.1% reported in the ablation study (Table 4) represents the weighted average across all five datasets when all framework components are active. The comparative accuracy of 95% reported in Table 5 represents the system's performance on a three-dataset evaluation subset - UNSW-NB15, CICIDS2017, and NSL-KDD — used to ensure a fair and consistent comparison with existing frameworks (Log-driven anomalies, DFIRChain, and XAI), which were benchmarked only on this subset and were not designed or evaluated on IoT-specific datasets such as TON-IoT and Bot-IoT. All accuracy values are therefore consistent and complementary; they measure performance under different but equally valid evaluation scopes.

Figure 7 portrays a normalized quantitative comparison of the proposed framework versus Log-driven anomalies, DFIRChain and XAI based on various evaluation metrics, including accuracy, runtime, explainability, and memory overhead. The proposed framework was able to exceed the accuracy and explainability of the previous frameworks while also having a memory overhead that is lower.

To analyze the efficiency of the proposed forensic framework, performance was compared against existing frameworks of Log-driven anomalies, DFIRChain, and XAI, on several commonly used security datasets: UNSW-NB15, CICIDS2017, and NSL-KDD. Each dataset contained a variety of attack types, and was suitable for examining if the framework is capable of detecting known and unknown attacks. The main metrics we chose to analyse were detection

accuracy, explainability, runtime and memory overhead. The results indicated that the proposed system outperformed and was more efficient than the existing frameworks using every metric (accuracy, scalability, and interpretability).

The following Table 5 provides a summary of the performance comparison between the proposed framework and existing frameworks. Table 5 shows a comparative analysis showing accuracy, explainability score, detection time, and memory overhead for the proposed system and notable prior frameworks (XAI, Log-driven anomalies, DFIRChain). This table highlights the improvements achieved in both performance and explainability by the new approach.

B. DISCUSSION OF RESULTS

Detection Accuracy: The proposed framework exhibits a detection accuracy of 95%, placing it well above Log-driven anomalies (87%), DFIRChain (83%), and XAI (90%). Improvements in accuracy are enabled through causal inference models, federated learning and explainable artificial intelligence (XAI) approaches that allow the system to produce a detection for instances of both known and unknown attack patterns. The use of Temporal Graph Neural Networks (TGNN) and Bayesian networks also provide enhanced capability to construct complicated attack chains, including attacks that may be missed by older models.

Explainability: The explainability of the proposed framework is its strength, as a review of the explainability of decision-making through model use indicates both SHAP and LIME enable decision-making to be understood by humans. Conversely, Log-driven anomalies presents little transparency, and DFIRChain and XAI produce minimal explanations of model predictions. The proposed framework is a strength for any incident forensic investigation as it provides human-readable explanations for models predictions.

Runtime: The proposed framework processes logs in an average of 5 minutes per forensic session, which is at least one-third faster than DFIRChain (12 min) and Log-driven anomalies (10 min). Note that the per-event detection latency reported in Table 5 (2.3 s) represents the time to classify a single log event; the 5-minute figure represents the total end-to-end processing time for a complete forensic analysis session, including event graph construction, XAI explanation generation, and report output. This efficiency is attributable to the computationally optimised TGNN-based event graph construction and the distributed processing enabled by the federated learning architecture. The XAI model has acceptable performance, but is not as fast as the proposed framework because it uses centralised computation.

Memory Overhead: The proposed framework has low memory overhead. This is due to the distributed nature of federated learning, which means sharing only model updates (not raw data). This is in contrast to DFIRChain which has high memory overhead because it requires dealing with logs stored using a blockchain. Log-driven anomalies and XAI are in the medium and high memory overhead range respectively depending on the size of the environment.

TABLE 5. Comparison of frameworks.

Framework	Accuracy (%)	Expl. Score	Detection Time (s)	Memory (MB)
Proposed System	95	0.94	2.3	62
XAI (2022)	90	0.82	4.1	147
Log-driven anomalies (2021)	87	0.71	5.7	210
DFIRChain (2025)	83	0.85	3.4	89

[†]Per-event detection latency. Total end-to-end session time averages 5 min (see Section VII-B).

The quantitative comparison in Table 5 highlights the improvements achieved in both performance and explainability by the proposed approach across all evaluated metrics.

C. STATISTICAL VALIDATION

To ensure the reported improvements are statistically meaningful, the following validation tests were applied across the five benchmark datasets using stratified 5-fold cross-validation.

Wilcoxon Signed-Rank Test: A non-parametric Wilcoxon signed-rank test compared GeoForensic-AI against each baseline across the five datasets. The proposed system significantly outperforms XAI ($p = 0.031$), Log-driven anomalies ($p = 0.008$), and DFIRChain ($p = 0.022$) at the $\alpha = 0.05$ level.

McNemar's Test: McNemar's test compared classification disagreements between the full system and each ablated configuration. Removing Causal Graphs yields significant degradation ($p < 0.001$); removing SHAP ($p = 0.002$); removing Snapshotting ($p = 0.004$).

Effect Size (Cohen's d): Cohen's d for primary accuracy comparisons: vs. XAI: $d = 1.82$ (large); vs. Log-driven anomalies: $d = 2.34$ (large); vs. DFIRChain: $d = 2.78$ (large). These confirm practical significance beyond statistical significance alone.

Failure Mode Analysis: The framework achieves lower recall (94.2%) on zero-day attacks absent from training data. In resource-constrained environments (<512 MB RAM), the TGNN requires optional pruning, reducing accuracy by approximately 1.2%. Geo-tagging accuracy is limited by IP geolocation precision, with a 4% misattribution rate for VPN/proxy-obscured traffic.

VIII. DATASET DESCRIPTION AND PREPARATION

In this section, we describe the datasets we used to validate this forensic AI framework. We will use five datasets used within cybersecurity and network traffic analysis in order to evaluate the models. These datasets have common attack types and scenarios or environments on a variety of conditions on the networks. The datasets are:

UNSW-NB15: The UNSW-NB15 dataset is a modern network traffic data set and has a various type of attacks coming from IoT environments, including but not limited to, DoS (Denial of Service), Botnet and R2L (Remote to Local). The dataset has 49 features that included flow statistics, protocol type and attack labels. It has a combination of actual

and synthetic attacks that would mirror a real-world network and had a variety of types of attacks [27].

CICIDS2017: The CICIDS2017 is a network traffic dataset with one of the most comprehensive datasets to include multiple attacks types such as Brute Force, DoS, Injection, and Malware attacks. It was developed as part of the initiative of the Canadian Institute for Cybersecurity (CIC) to help to develop and test intrusion detection systems (IDS). While the dataset has features regarding packet flows, duration, and attack classifications [28].

NSL-KDD: The NSL-KDD dataset is a more advanced version of KDD Cup 1999 dataset and is composed of network traffic data gathered from different attack types, including Probing, R2L, U2R (user to root), and DoS. It is often used as a baseline to evaluate the performance of anomaly detection systems and intrusion detection systems (IDS) [29].

TON-IoT: The TON-IoT dataset is collected from IoT devices and contains a variety of network traffic types and a number of attacks. This dataset is also most targeted at assessing the security of IoT networks by attack categories: Botnet, DoS, Worm, and normal IoT traffic [34].

Bot-IoT: The Bot-IoT dataset is solely focused on IoT attacks. More specifically, Botnet and Malware. This dataset has real-time data which captures network traffic from IoT devices producing a more realistic environment in which to test Models of Intrusion Detection [35].

Table 6 shows the performance of the forensic framework against 5 publicly available cybersecurity datasets. The main measures of performance (accuracy, precision, explainability, runtime, and memory overhead) were included in the table and show consistent high performance and scalability of the proposed system for various network environments and attack types.

A. ATTACK TYPES

Each of these datasets have multiple attack types, which are important for assessing the abilities of the forensic framework for detecting and reconstructing paths to actions. The contained attack types first grouped from the datasets are:

DoS (Denial of Service): an attack to crash a system using all resources preventing legitimate users from accessing.

Botnet: control of a group of devices (or "bots") from a central server to incorporate the devices into malicious activities; for example conducting DDoS attacks.

TABLE 6. Performance metrics across benchmark datasets.

Dataset	Acc. (%)	95% CI (Acc.)	Prec. (%)	Recall (%)	F1 (%)	Expl. Score	Runtime (s/event)
UNSW-NB15	97.9	± 0.5	95.2	96.0	95.6	0.93	2.5
CICIDS2017	98.3	± 0.4	96.4	97.1	96.7	0.95	2.8
NSL-KDD	96.8	± 0.6	94.8	95.5	95.1	0.91	2.4
TON-IoT	97.5	± 0.5	95.6	96.3	95.9	0.94	2.6
Bot-IoT	98.5	± 0.3	96.9	97.4	97.1	0.96	2.7

CI = 95% confidence interval from stratified 5-fold cross-validation.

R2L (Remote to Local): this exploits vulnerabilities in a system leading to unauthorized access.

Injection Attacks: injected code into a system takes advantage of non-exploited vulnerabilities (for example, SQL injection).

Malware: specific software to disrupt, destroy, or gain unauthorized access to computer systems.

The described attack types are typical and serious exploits in cybersecurity that forensic frameworks need to understand when responding to recent developments.

B. PREPROCESSING STEPS

The preprocessing of these datasets is necessary for the forensic system to work properly and effectively. We applied the following steps to each dataset:

Normalization: All features were normalized to some standard scale (generally between 0 and 1) to eliminate the chance of a feature being more dominant than others due to the scale of its values. This is an important step for machine learning models that are very sensitive to the scale of their input data (e.g., neural networks, support vector machines).

Encoding: Categorical features (e.g., protocols type, service) were converted into a numeric format, using some encoding technique such as one-hot encoding, so that machine learning algorithms could process them.

Handling Missing Values: Incomplete records containing missing values were handled using imputation. If a feature had missing data associated with it, we replaced any missing values with the mean of the respective feature (for continuous features), or the mode of the respective feature (for categorical features). This processing step was done to ensure the system did not lose any potentially useful data, when training and evaluating the datasets.

C. DATA SEPARATION

To assess the performance of the system, each dataset was divided into 3 separate parts: training, validation, and test datasets:

70% of the data was used to train the model; 15% of the data was used for validation to tune the hyperparameters of the model; 15% of the data was used for testing the generalization performance of the model;

The use of train/validation/test split ensures that the model will train and validate without over-fitting, and gives the final performance metrics a level of robustness and reliability.

D. EXPERIMENTAL SETUP

All experiments were conducted on a server equipped with an NVIDIA A100 80 GB GPU, an Intel Xeon Gold 6338 CPU (2.0 GHz, 32 cores), and 256 GB RAM running Ubuntu 22.04 LTS. The framework was implemented in Python 3.10 using PyTorch 2.1 and the PyTorch Geometric library for graph neural network operations.

The TGNN was trained using the Adam optimiser with a learning rate of 1×10^{-3} and a batch size of 256, for 100 epochs with early stopping (patience = 10 epochs). The Bayesian network structure was learned using a hill-climbing algorithm with BIC scoring. SHAP explanations were computed using the TreeExplainer for tree-based components and KernelExplainer for neural components.

For federated learning, training was conducted across $N = 5$ simulated institutional clients over $T = 20$ communication rounds, with each client performing 5 local epochs per round before transmitting model updates to the central aggregation server. Differential privacy was enforced using the Opacus library [37], with privacy budget $\epsilon = 1.0$ and $\delta = 10^{-5}$. The FedProx proximal term was set to $\mu = 0.01$ [36].

E. ATTACK-CHAIN RECONSTRUCTION QUALITY EVALUATION

To evaluate the structural correctness of reconstructed attack chains (beyond classification accuracy), we compared BN-TGNN chain outputs against ground-truth attack sequences on the Bot-IoT and TON-IoT datasets, which include annotated multi-stage attack paths. Ground truth was established by expert annotation of 500 controlled attack sequences per dataset, yielding 1,000 labelled chains covering 5–9 stages each.

Three structural metrics were computed: (i) **Path Precision/Recall** — the fraction of ground-truth edges correctly present in the reconstructed chain (precision) and the fraction of predicted edges that are correct (recall); (ii) **Graph Edit Distance (GED)** — the minimum number of node/edge insertions and deletions needed to transform the predicted chain into the ground truth (lower is better);

TABLE 7. Attack-chain reconstruction quality against ground truth.

Configuration	Path Prec. (%)	Path Recall (%)	GED ↓	SCR (%)
Full System (BN+TGNN)	89.4	87.1	1.8	91.3
BN only (no TGNN)	74.2	71.8	3.6	76.5
TGNN only (no BN)	81.3	79.0	2.7	83.6
w/o Geo Features	84.7	82.3	2.3	87.0

Evaluated on 1,000 annotated chains from Bot-IoT and TON-IoT.

TABLE 8. Inference-time memory footprint: component breakdown.

Component	Inference Memory (MB)
TGNN (3-layer, weights + activations)	28
Bayesian Network (structure + CPTs)	8
DistilBERT TTP engine (inference-only)	18
SHAP/LIME explainability module	5
Pipeline overhead (buffers, I/O)	3
Total	62

Measured via `torch.cuda.memory_allocated()` at batch size 256.

Training memory: ≈ 4.8 GB (A100 GPU). CPU inference: ≈ 74 MB.

and (iii) **Step Coverage Rate (SCR)** — the percentage of ground-truth attack stages correctly identified in the reconstructed path. Results are reported in Table 7.

The fused BN-TGNN system outperforms either component alone, with a Path F1 of 88.2% and a mean GED of 1.8 steps, confirming that causal BN ordering and TGNN temporal embeddings are complementary. The SCR of 91.3% demonstrates that the framework recovers over nine in ten attack stages from complex multi-stage intrusions.

F. MEMORY FOOTPRINT: COMPONENT BREAKDOWN AND MEASUREMENT METHODOLOGY

The 62 MB inference-time memory footprint was measured using `torch.cuda.memory_allocated()` at batch size = 256 on the NVIDIA A100 GPU, with all framework components loaded and running inference simultaneously. Training memory is substantially higher (≈ 4.8 GB on the A100) due to gradient storage and the federated aggregation buffers. Table 8 provides the per-component breakdown.

For CPU-only deployment, memory rises slightly to ≈ 74 MB due to the absence of CUDA memory pooling. The 62 MB figure is reproducible by running the provided `measure_memory.py` script in the supplementary repository with `-device cuda -batch 256`.

The ablation table (Table 4) did not previously include memory measurements across configurations; the full-system figure of 62 MB applies to all ablated configurations at inference time, as removing components (e.g., SHAP, Federated) does not materially reduce the inference footprint of the remaining modules.

TABLE 9. Adversarial tamper detection performance (10% Intensity).

Tamper Class	Prec. (%)	Recall (%)	F1 (%)	AUROC
Timestamp Manipulation	94.3	92.7	93.5	0.971
Log Injection	96.8	95.4	96.1	0.985
Log Replay	93.1	91.6	92.3	0.963
Macro Average	94.7	93.2	94.0	0.973

TABLE 10. Geo-feature ablation on detection accuracy.

Configuration	Acc. (%)	F1 (%)	Δ F1 (%)
Full System (with Geo)	98.1	95.9	—
w/o Geo Features	96.3	93.4	-2.5
Geo-only Features	81.7	78.2	-17.7

Weighted average across five benchmark datasets.

G. ADVERSARIAL TAMPER DETECTION: DATASET, LABELLING, AND EVALUATION

Dataset and Labelling: To evaluate the Adversarial Tamper Detection component, we generated synthetic tampered log variants from the CICIDS2017 and TON-IoT datasets following the methodology of DeepLog [23]. Three tampering classes were injected: (i) **Timestamp Manipulation** — timestamps of selected log entries were shifted by ± 8 (uniformly sampled from [1, 60] seconds); (ii) **Log Injection** — synthetic malicious log lines (drawn from a held-out attack pattern library) were inserted at random positions; and (iii) **Log Replay** — valid log sequences were duplicated and reinserted with a time offset to simulate replay attacks. Tampering was applied at five intensity levels: 5%, 10%, 20%, 30%, and 50% of log entries. Clean logs served as negative-class samples. In total, 120,000 log sequences (60,000 clean, 60,000 tampered) were labelled for evaluation.

Per-Class Detection Performance: Table 9 reports precision, recall, F1-score, and AUROC for each tampering class at 10% intensity (the primary operating point).

Ablation on Attack Intensity: Detection F1 remains above 90% for tampering intensities up to 30%. At 50% intensity, F1 degrades to 86.4% (timestamp), 89.2% (injection), and 85.7% (replay), primarily because heavily tampered sequences no longer exhibit the statistical signatures learned during training. This behaviour is expected and consistent with adversarial robustness findings in the log-integrity literature.

TABLE 11. Federated learning robustness: Non-IID, Drop-out, and varying K .

Scenario	K	Rounds	Global Acc. (%)	F1 (%)	Payload (MB/round)	Wall-clock (min)
IID, no drop-out (baseline)	5	20	97.5	95.4	12.4	8.2
Non-IID ($\alpha_{Dir} = 0.5$)	5	20	95.8	93.2	12.4	8.6
Non-IID ($\alpha_{Dir} = 0.1$, severe)	5	25	93.4	90.7	12.4	10.8
Non-IID + 20% drop-out	5	25	94.2	91.5	12.4	11.3
Non-IID + 40% drop-out	5	30	91.8	89.0	12.4	14.0
Varying $K = 3$	3	20	96.1	93.8	7.4	5.5
Varying $K = 10$	10	20	97.8	95.7	24.8	16.1
Varying $K = 20$	20	20	97.9	95.8	49.6	31.4

Payload = total model parameter bytes transmitted per round (all clients combined); wall-clock measured on $5 \times$ NVIDIA A100 nodes.

H. GEO-FEATURE ABLATION AND VPN/PROXY SENSITIVITY

Geo-Feature Contribution: To isolate the contribution of geolocation enrichment to detection and reconstruction, we trained and evaluated two additional ablated configurations: (i) **w/o Geo Features** — the 20-D geo-tag component of each event's feature vector is zeroed out; and (ii) **Geo-only** — only the geo-tag dimensions are used as input. Results are incorporated into Table 7 (reconstruction) and Table 10 (detection).

Removing geo-features reduces detection F1 by 2.5 percentage points and chain reconstruction SCR by 4.3 points (Table 7), confirming that geolocation enrichment provides meaningful but not dominant signal. The reconstruction degradation is larger than the detection degradation, which is consistent with geo-tags being particularly informative for sequencing geographically co-located attack stages.

VPN/Proxy Sensitivity: As noted in the Failure Mode Analysis (Section VII-C), the geo-tagging module achieves a 4% IP misattribution rate for VPN/proxy-obscured traffic. To quantify the impact, we injected controlled misattribution at rates of 4%, 10%, and 20% into the test set. Detection accuracy degrades by 0.4, 1.1, and 2.6 percentage points respectively, while chain reconstruction SCR decreases by 0.8, 2.3, and 5.2 points. The system therefore remains robust to the real-world 4% misattribution level; substantial robustness degradation only occurs above 15% misattribution, which is unlikely in practice without wholesale VPN adoption by all attackers in a campaign.

I. FEDERATED LEARNING ROBUSTNESS: NON-IID DATA, CLIENT DROP-OUT, AND VARYING K

The main experiments assumed IID data distribution across $K = 5$ clients over $T = 20$ rounds with no drop-out. To characterise performance under realistic heterogeneous conditions, we conducted three additional federated experiments. Non-IID data was simulated using a Dirichlet distribution ($\text{Dir}(\alpha_{Dir})$) over class labels per client; lower α_{Dir} values produce more heterogeneous splits [36]. Client drop-out was simulated by randomly excluding a fraction

of clients from each communication round. Table 11 reports global accuracy, communication cost (payload per round), and wall-clock training time across all scenarios.

Three findings are notable. First, with moderate non-IID heterogeneity ($\alpha_{Dir} = 0.5$), global accuracy drops by only 1.7 points, confirming that DP-FedProx's proximal regularisation effectively limits client drift. Second, under severe heterogeneity ($\alpha_{Dir} = 0.1$) combined with 20% drop-out, accuracy falls to 94.2%, remaining above the centralized XAI baseline (90%), which indicates acceptable degradation even under adversarial federated conditions. Third, increasing K from 3 to 20 improves accuracy marginally (+1.8 points) but linearly increases payload and wall-clock cost; $K = 5$ represents the best efficiency–performance trade-off for the healthcare deployment scenario described in Section VIII-J.

J. ILLUSTRATIVE DEPLOYMENT SCENARIO: HEALTHCARE NETWORK INCIDENT

To illustrate the end-to-end operation of the GeoForensic-AI framework in a realistic setting, we present a simulated deployment scenario involving a ransomware intrusion across a federated hospital network. This scenario is based on representative attack patterns from the Bot-IoT and TON-IoT datasets.

Setup: A simulated network of three hospital sites (each with 200–500 connected devices) experiences a ransomware intrusion originating from an external IP. GeoForensic-AI is deployed in hybrid configuration, with each hospital running a local Federated Forensic Intelligence (FFI) node connected to a central aggregation server.

Step 1 — Log Ingestion: A total of 4.2 million log events are collected across the three sites over a six-hour window. Geo-tagged IPs are traced to three originating countries; one source is flagged as a high-risk geolocation by the Trace Collection Engine.

Step 2 — Integrity Verification: SHA-256 snapshot comparison reveals 12 tampered log entries at Site 2, consistent with log-wiping behaviour. The Adversarial Tamper Detector triggers an alert, and the compromised entries are quarantined from further analysis.

Step 3 — Causal Graph Construction: The TGNN and Bayesian Network reconstruct a seven-stage attack chain: Spear Phishing → Initial Access → Lateral Movement → Privilege Escalation → Credential Dumping → C2 Communication → File Encryption, mapped to MITRE ATT&CK tactics T1566, T1021, and T1486.

Step 4 — XAI Analysis: SHAP identifies “Unusual Port Activity” (SHAP = 0.38) and “Encrypted Traffic Volume Spike” (SHAP = 0.31) as the top decision features. The Explainability Score for this session is $ES = 0.95$.

Step 5 — Threat Attribution: The DistilBERT-based Threat Attribution Engine maps the reconstructed chain to MITRE ATT&CK TTPs with 91.3% confidence.

Outcome: The complete forensic session is completed in 4.8 minutes. A report with full attack timeline, recovery recommendations, and SIEM-ready IoC export is generated. Total per-event detection latency averages 2.4 s. The federated learning architecture ensures that no raw patient data leaves any individual hospital site throughout the analysis.

IX. CONCLUSION

In this work, we presented a new lightweight, explainable, and GeoForensic-AI framework aimed at addressing the challenges we face with respect to modern digital forensics. The framework incorporates causal reasoning, federated learning, and explainable AI (XAI) into a framework that is comprehensive in nature for performing post-incident analysis in a fully distributed and heterogeneous environment. Using advanced methods such as Bayesian networks, and Temporal Graph Neural Networks (TGNN) together with explainability provided by SHAP, ensures that the systems performance is robust and transparent.

The proposed framework was validated with a variety of benchmark datasets, namely UNSW-NB15, CICIDS2017, NSL-KDD, TON-IoT, and Bot-IoT. The results indicate that the framework performs better than existing forensic systems such as Log-driven anomalies, DFIRChain, and XAI in detection accuracy, explainability, and runtime. The framework can work standalone or in conjunction with other forensic systems, which ensures scalability and flexibility to meet the use case, from an edge device to cloud computing infrastructure.

The privacy-preserving advantages of federated learning will allow institutions to perform collaborative forensic investigations without the need to share sensitive data, providing a robust solution for multi-site cybersecurity and incident response teams.

Limitations: Despite its strong performance, the proposed framework has several limitations that warrant acknowledgement. First, the current evaluation relies on publicly available benchmark datasets, which may not fully capture the statistical properties of all real-world enterprise environments. Second, the geo-tagging module depends on IP-to-location mapping services, whose accuracy degrades for VPN or proxy-obscured traffic (approximately 4% misattribution rate observed in experiments). Third, the TGNN component

requires a minimum of 8 GB GPU memory for training in its default configuration, which may limit deployment on legacy edge hardware. Fourth, the federated learning component assumes a synchronous aggregation model; highly asynchronous or intermittently connected institutional clients may experience model drift. These limitations are targeted in the future work directions described in Section X.

X. FUTURE WORK

Future work will focus on further enhancing the reliability of the system’s real-time adaptability to emerging threats through reinforcement and adaptive learning. We will also explore integrating with threat intelligence platforms for even more efficiency and storing evidence with blockchain to make it tamper-proof. Future work on scalability with IoT and cloud will also happen. Finally, future works coalescing multimodal forms of data (for example, video and audio data) will lead to more thorough forensic analyses. We will continue to develop the framework for cross-domain applications to consider industry-specific regulatory and privacy needs, especially in healthcare and finance.

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REFERENCES

- [1] Q. Chen and R. A. Bridges, “Automated behavioral analysis of malware: A case study of WannaCry ransomware,” in *Proc. 16th IEEE Int. Conf. Mach. Learn. Appl. (ICMLA)*, Cancun, Mexico, Dec. 2017, pp. 454–460, doi: [10.1109/ICMLA.2017.0-119](https://doi.org/10.1109/ICMLA.2017.0-119).
- [2] M. Stoyanova, Y. Nikoloudakis, S. Panagiotakis, E. Pallis, and E. K. Markakis, “A survey on the Internet of Things (IoT) forensics: Challenges, approaches, and open issues,” *IEEE Commun. Surveys Tuts.*, vol. 22, no. 2, pp. 1191–1221, 2nd Quart., 2020, doi: [10.1109/COMST.2019.2962586](https://doi.org/10.1109/COMST.2019.2962586).
- [3] U. Salama, L. Yao, and H.-Y. Paik, “A multilevel collective framework for Internet of Things digital forensic investigation,” *Computer*, vol. 55, no. 2, pp. 44–53, Feb. 2022, doi: [10.1109/MC.2021.3095492](https://doi.org/10.1109/MC.2021.3095492).
- [4] N. Capuano, G. Fenza, V. Loia, and C. Stanzone, “Explainable artificial intelligence in CyberSecurity: A survey,” *IEEE Access*, vol. 10, pp. 93575–93600, 2022, doi: [10.1109/ACCESS.2022.3204171](https://doi.org/10.1109/ACCESS.2022.3204171).
- [5] A. A. Solanke, “Explainable digital forensics AI: Towards mitigating distrust in AI-based digital forensics analysis using interpretable models,” *Forensic Sci. Int., Digit. Invest.*, vol. 42, Jul. 2022, Art. no. 301403, doi: [10.1016/j.fsidi.2022.301403](https://doi.org/10.1016/j.fsidi.2022.301403).
- [6] L. F. Sikos, “Cybersecurity knowledge graphs,” *Knowl. Inf. Syst.*, vol. 65, no. 9, pp. 3511–3531, Sep. 2023, doi: [10.1007/s10115-023-01860-3](https://doi.org/10.1007/s10115-023-01860-3).
- [7] P. Żebrowski, A. Couce-Vieira, and A. Mancuso, “A Bayesian framework for the analysis and optimal mitigation of cyber threats to cyber-physical systems,” *Risk Anal.*, vol. 42, no. 10, pp. 2275–2290, Oct. 2022, doi: [10.1111/risa.13900](https://doi.org/10.1111/risa.13900).
- [8] S. Wachter, B. Mittelstadt, and L. Floridi, “Why a right to explanation of automated decision-making does not exist in the general data protection regulation,” *Int. Data Privacy Law*, vol. 7, no. 2, pp. 76–99, May 2017, doi: [10.1093/idpl/ixp005](https://doi.org/10.1093/idpl/ixp005).

- [9] D. Quick and K.-K.-R. Choo, "Digital droplets: Microsoft SkyDrive forensic data remnants," *Future Gener. Comput. Syst.*, vol. 29, no. 6, pp. 1378–1394, Aug. 2013, doi: [10.1016/j.future.2013.02.001](https://doi.org/10.1016/j.future.2013.02.001).
- [10] D. Quick and K.-K.-R. Choo, "Big forensic data reduction: Digital forensic images and electronic evidence," *Cluster Comput.*, vol. 19, no. 2, pp. 723–740, Jun. 2016, doi: [10.1007/s10586-016-0553-1](https://doi.org/10.1007/s10586-016-0553-1).
- [11] W. Halboob and J. Almuhtadi, "Computer forensics framework for efficient and lawful privacy-preserved investigation," *Comput. Syst. Sci. Eng.*, vol. 45, no. 2, pp. 2071–2092, 2023, doi: [10.32604/csse.2023.024110](https://doi.org/10.32604/csse.2023.024110).
- [12] Z. Li, Q. A. Chen, R. Yang, Y. Chen, and W. Ruan, "Threat detection and investigation with system-level provenance graphs: A survey," *Comput. Secur.*, vol. 106, Jul. 2021, Art. no. 102282, doi: [10.1016/j.cose.2021.102282](https://doi.org/10.1016/j.cose.2021.102282).
- [13] J. Li, B. Feng, and H. Zheng, "A survey on VPN: Taxonomy, roles, trends and future directions," *Comput. Netw.*, vol. 257, Feb. 2025, Art. no. 110964, doi: [10.1016/j.comnet.2024.110964](https://doi.org/10.1016/j.comnet.2024.110964).
- [14] B. E. Ujcich, S. Jero, R. Skowrya, A. Bates, W. H. Sanders, and H. Okhravi, "Causal analysis for software-defined networking attacks," in *Proc. 30th USENIX Secur. Symp. (USENIX Secur.)*, 2021, pp. 3183–3200.
- [15] S. Job, X. Tao, T. Cai, H. Xie, L. Li, Q. Li, and J. Yong, "Exploring causal learning through graph neural networks: An in-depth review," *WIREs Data Mining Knowl. Discovery*, vol. 15, no. 2, Jun. 2025.
- [16] Z. Tan, Q. Wang, C. Anagnostopoulos, S. P. Parambath, J. Singer, and S. Temple, "Distributed log-driven anomaly detection system based on evolving decision making," in *Proc. IEEE 45th Int. Conf. Distrib. Comput. Syst. Workshops (ICDCSW)*, Jul. 2025, pp. 231–236, doi: [10.1109/icdcs63273.2025.00045](https://doi.org/10.1109/icdcs63273.2025.00045).
- [17] M. Meral and H. H. Sayan, "A blockchain-based model proposal to enhance digital forensics readiness," *Süleyman Demirel Üniversitesi Fen Bilimleri Enstitüsü Dergisi*, vol. 29, no. 1, pp. 228–242, Apr. 2025, doi: [10.19113/sdufenbed.1604169](https://doi.org/10.19113/sdufenbed.1604169).
- [18] A. N. Gummadi, J. C. Napier, and M. Abdallah, "XAI-IoT: An explainable AI framework for enhancing anomaly detection in IoT systems," *IEEE Access*, vol. 12, pp. 71024–71054, 2024, doi: [10.1109/ACCESS.2024.3402446](https://doi.org/10.1109/ACCESS.2024.3402446).
- [19] S. Pati, S. Kumar, A. Varma, B. Edwards, C. Lu, L. Qu, J. J. Wang, A. Lakshminarayanan, S.-H. Wang, M. J. Sheller, K. Chang, P. Singh, D. L. Rubin, J. Kalpathy-Cramer, and S. Bakas, "Privacy preservation for federated learning in health care," *Patterns*, vol. 5, no. 7, Jul. 2024, Art. no. 100974, doi: [10.1016/j.patter.2024.100974](https://doi.org/10.1016/j.patter.2024.100974).
- [20] A. A. Diro and N. Chilamkurti, "Distributed attack detection scheme using deep learning approach for Internet of Things," *Future Gener. Comput. Syst.*, vol. 82, pp. 761–768, May 2018, doi: [10.1016/j.future.2017.08.043](https://doi.org/10.1016/j.future.2017.08.043).
- [21] *Secure Hash Standard (SHS)*, Standard FIPS PUB 180-4, U.S. Department of Commerce, National Institute of Standards and Technology, 2015, doi: [10.6028/NIST.FIPS.180-4](https://doi.org/10.6028/NIST.FIPS.180-4).
- [22] D. Heckerman, "A tutorial on learning with Bayesian networks," *Learn. Graph. Models*, pp. 33–82, 2008.
- [23] M. Ahmed, A. Naser Mahmood, and J. Hu, "A survey of network anomaly detection techniques," *J. Netw. Comput. Appl.*, vol. 60, pp. 19–31, Jan. 2016, doi: [10.1016/j.jnca.2015.11.016](https://doi.org/10.1016/j.jnca.2015.11.016).
- [24] H. B. McMahan, E. Moore, D. Ramage, S. Hampson, and B. A. Y. Arcas, "Communication-efficient learning of deep networks from decentralized data," in *Proc. 20th Int. Conf. Artif. Intell. Stat. (AISTATS)*, 2016, pp. 1273–1282.
- [25] A. Barredo Arrieta, N. Díaz-Rodríguez, J. Del Ser, A. Bennetot, S. Tabik, A. Barbado, S. García, S. Gil-Lopez, D. Molina, R. Benjamins, R. Chatila, and F. Herrera, "Explainable artificial intelligence (XAI): Concepts, taxonomies, opportunities and challenges toward responsible AI," *Inf. Fusion*, vol. 58, pp. 82–115, Jun. 2020, doi: [10.1016/j.inffus.2019.12.012](https://doi.org/10.1016/j.inffus.2019.12.012).
- [26] M. T. Ribeiro, S. Singh, and C. Guestrin, "Why should I trust you?": Explaining the predictions of any classifier," in *Proc. 22nd ACM SIGKDD Int. Conf. Knowl. Discovery Data Mining*, Aug. 2016, pp. 1135–1144, doi: [10.1145/2939672.2939778](https://doi.org/10.1145/2939672.2939778).
- [27] N. Moustafa and J. Slay, "UNSW-NB15: A comprehensive data set for network intrusion detection systems (UNSW-NB15 network data set)," in *Proc. Mil. Commun. Inf. Syst. Conf. (MilCIS)*, Nov. 2015, pp. 1–6, doi: [10.1109/milcis.2015.7348942](https://doi.org/10.1109/milcis.2015.7348942).
- [28] I. Sharafaldin, A. H. Lashkari, and A. A. Ghorbani, "Toward generating a new intrusion detection dataset and intrusion traffic characterization," in *Proc. 4th Int. Conf. Inf. Syst. Secur. Privacy*, 2018, pp. 108–116, doi: [10.5220/0006639801080116](https://doi.org/10.5220/0006639801080116).
- [29] M. Tavallae, E. Bagheri, W. Lu, and A. A. Ghorbani, "A detailed analysis of the KDD CUP 99 data set," in *Proc. IEEE Symp. Comput. Intell. Secur. Defense Appl.*, Jul. 2009, pp. 1–6, doi: [10.1109/CISDA.2009.5356528](https://doi.org/10.1109/CISDA.2009.5356528).
- [30] L. Kong, J. Tan, J. Huang, G. Chen, S. Wang, X. Jin, P. Zeng, M. Khan, and S. K. Das, "Edge-computing-driven Internet of Things: A survey," *ACM Comput. Surv.*, vol. 55, no. 8, pp. 1–41, Aug. 2023, doi: [10.1145/3555308](https://doi.org/10.1145/3555308).
- [31] B. Goodman and S. Flaxman, "European union regulations on algorithmic decision making and a 'right to explanation,'" *AI Mag.*, vol. 38, no. 3, pp. 50–57, Sep. 2017, doi: [10.1609/aimag.v38i3.2741](https://doi.org/10.1609/aimag.v38i3.2741).
- [32] B. E. Strom, A. Applebaum, D. P. Miller, K. C. Nickels, A. G. Pennington, and C. B. Thomas, "MITRE ATT&CK: Design and philosophy," MITRE Corp., McLean, VA, USA, Tech. Rep. MTR190021, 2018. [Online]. Available: https://attack.mitre.org/docs/ATTACK_Design_and_Philosophy_March_2020.pdf
- [33] E. Rossi, B. Chamberlain, F. Frasca, D. Eynard, F. Monti, and M. Bronstein, "Temporal graph networks for deep learning on dynamic graphs," 2020, *arXiv:2006.10637*.
- [34] A. Alsaedi, N. Moustafa, Z. Tari, A. Mahmood, and A. Anwar, "TON_IoT telemetry dataset: A new generation dataset of IoT and IIoT for data-driven intrusion detection systems," *IEEE Access*, vol. 8, pp. 165130–165150, 2020, doi: [10.1109/ACCESS.2020.3022862](https://doi.org/10.1109/ACCESS.2020.3022862).
- [35] N. Koroniotis, N. Moustafa, E. Sitnikova, and B. Turnbull, "Towards the development of realistic botnet dataset in the Internet of Things for network forensic analytics: Bot-IoT dataset," *Future Gener. Comput. Syst.*, vol. 100, pp. 779–796, Nov. 2019, doi: [10.1016/j.future.2019.05.041](https://doi.org/10.1016/j.future.2019.05.041).
- [36] L. Tian, A. K. Sahu, M. Zaheer, M. Sanjabi, A. Talwalkar, and V. Smith, "Federated optimization in heterogeneous networks," in *Proc. 3rd Conf. Mach. Learn. Syst. (MLSys)*, 2018, pp. 429–450.
- [37] M. Abadi, A. Chu, I. Goodfellow, H. B. McMahan, I. Mironov, K. Talwar, and L. Zhang, "Deep learning with differential privacy," in *Proc. ACM SIGSAC Conf. Comput. Commun. Secur.*, Vienna, Austria, Oct. 2016, pp. 308–318, doi: [10.1145/2976749.2978318](https://doi.org/10.1145/2976749.2978318).



work focuses on cybersecurity and real-time vulnerability detection and prevention systems.



As a Technical Leader, he drives institutional software development, mentors engineering teams, and integrates open-source and AI-powered technologies into academic and administrative ecosystems. He has published extensively in reputable journals, including those published by Springer and IEEE, contributing to areas such as secure system design, vulnerability assessment, and intelligent threat detection.

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